AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2018

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

AKEB May 2018 Test for Teaching & Learning Only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 9, 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

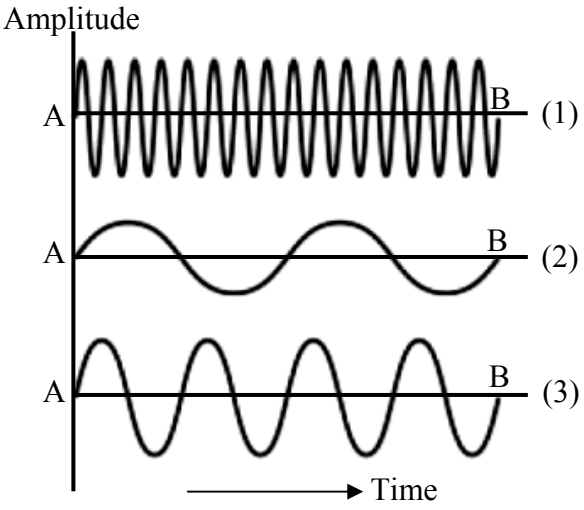
   Read each question carefully.
   Use a black pointer to write your answers.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

The given diagram shows three waves covering the same distance from points (A) to (B) in 2 seconds.



a. Write the frequency of any TWO waves. (2 Marks)

b. Which of the labelled waves, (1), (2) or (3), has the greatest speed and why? (2 Marks)

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Q.2. (Total 3 Marks)

Identify the following as 'True' OR 'False' and give a reason to justify any ONE of your answers.

a. Light travels slower through glass than empty space. _____ (1 Mark)

b. When light travels from glass to air, its path bends towards the normal. _____ (1 Mark)

Justification: (1 Mark)

_____

_____

_____

_____

Q.3. (Total 2 Marks)

Mention TWO conditions for the occurrence of total internal reflection.

_____

_____

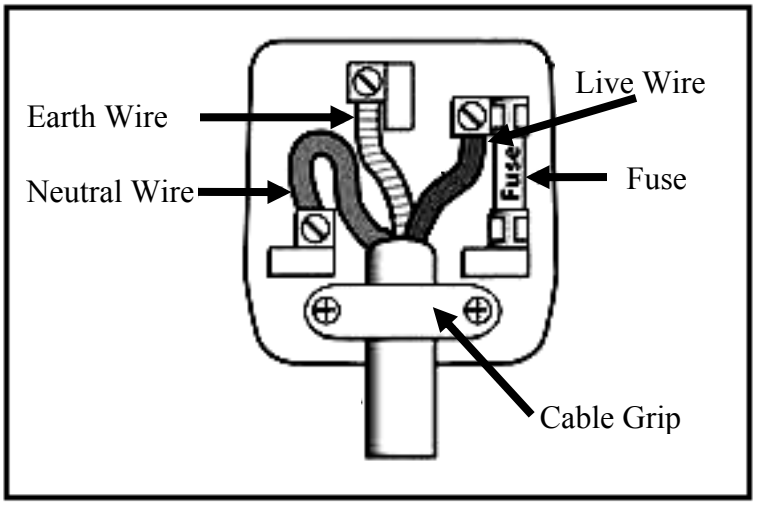
_____

_____

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Q.4. (Total 3 Marks)

The given figure shows a power plug.



a. State the function of the earth wire shown in the given figure. (1 Mark)

_____

_____

b. Describe the working of a fuse in the power plug. (1 Mark)

_____

_____

c. What would happen when one touches a power plug with wet hands or when one is barefoot? (1 Mark)

_____

_____

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Q.5. (Total 3 Marks)

Define the following:

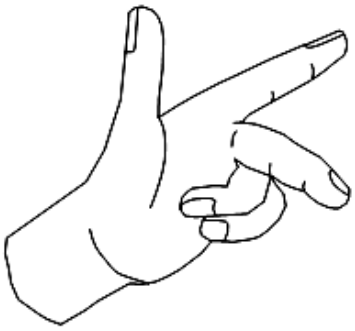
a. Electric field (1 Mark)

b. Electric field intensity (1 Mark)

c. Electrostatic potential (1 Mark)

Q.6. (Total 3 Marks)

The given figure shows the direction of the magnetic field, the current and the force on a conductor according to Fleming's left hand rule.



Match the items in column I with the items in column II using arrows (⟶).

Fleming's Left Hand Rule		
Column I	**Arrow**	**Column II**
The thumb		The direction of the current.
The index finger		The direction of the electric field.
The middle finger		The direction of the magnetic field.
		The direction of the force acting on a conductor.

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Q.7. (Total 3 Marks)

Mention any THREE benefits of transmitting information and data through optical fibre.

Q.8. (Total 4 Marks)

Reaction # 1:

$^{218}Po_{84} \; - \; ^{4}He_{2} \; = \; ^{214}Pb_{82}$

Reaction # 2:

$^{214}Pb_{82} \; - \; ^{0}e_{-1} \; = \; ^{214}Bi_{83}$

a. Which type of nuclear radiation causes the following changes? (2 Marks)

i. Conversion of polonium into lead in reaction # 1

ii. Conversion of lead into bismuth in reaction # 2

b. Give a reason to support your answer for (ii) of part (a). (2 Marks)

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Q.9. (Total 5 Marks)

EITHER

a. Describe any FIVE differences between mechanical and electromagnetic waves. (5 Marks)

OR

b.

i. Define the term 'echo'. (1 Mark)

ii. 'When an empty room starts to get filled with people, the echo in the room decreases.'

Elaborate the given statement. (4 Marks)

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Q.10. (Total 5 Marks)

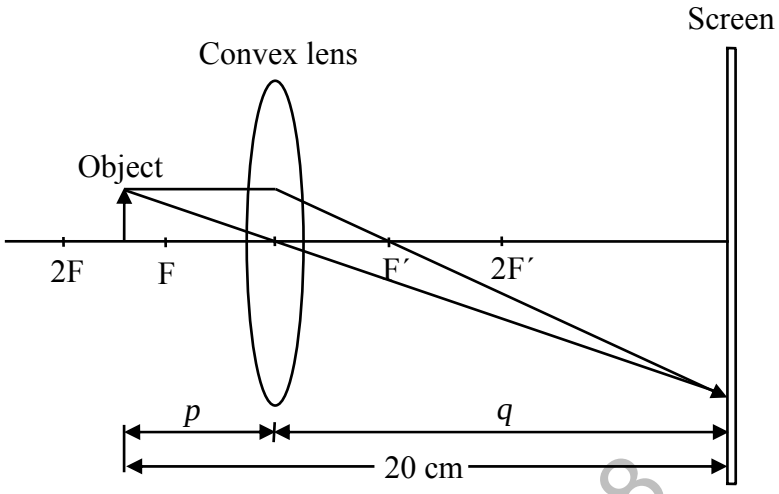
EITHER

a. Draw a well labelled ray diagram to illustrate the formation of an enlarged and virtual image through a converging mirror.

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OR

b. Consider the given ray diagram.



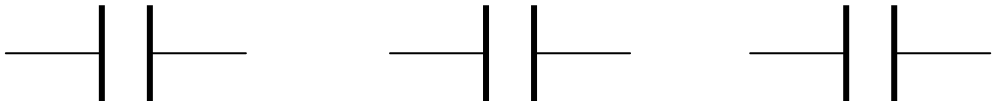
Calculate the value of focal length (f), if (q = 4p).

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Q.11. (Total 5 Marks)

EITHER

a. Three capacitors of capacitance (C_1), (C_2) and (C_3) are shown in the given figures.



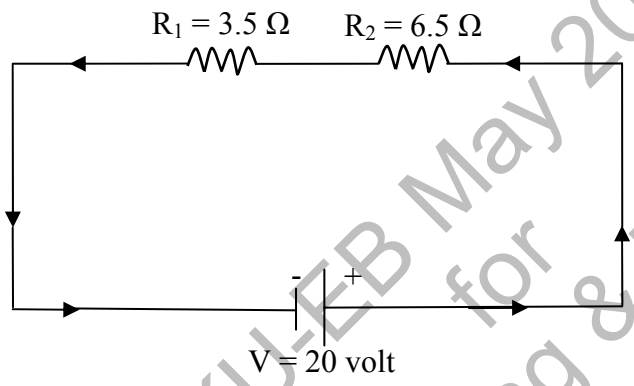
$C_1 = 2\ \mu F$ $C_2 = 4\ \mu F$ $C_3 = 8\ \mu F$

i. Draw a circuit diagram to show the parallel combination of the given capacitors with a power source. (3 Marks)

ii. Calculate the effective (equivalent) capacitance of the given capacitors in series combination. (2 Marks)

OR

b. The given circuit diagram shows the connection of two different resistors with a battery.



Calculate the following:

i. Effective (equivalent) resistance (3 Marks)

ii. Amount of current (2 Marks)

(**Note**: Provide your answers in S.I. units.)

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END OF PAPER

Please use this page for rough work

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